

Optimizing Random Forest Models for Stock Market Prediction with Hyperparameter Analysis

Yicong Li
College of Art and Science
New York University
New York, USA
yl8968@nyu.edu

Abstract

The forecasting of financial markets is an essential practice in the assessment of risk in financial transactions and investments. Traditional statistical techniques often suffer from various issues while analyzing such financial data and the evolving structure of market data. Random Forest (RF) is one of the most powerful machine learning methods to address the issues due to its ability to handle high-dimensional features and reduce overfitting. This paper assesses the effects of the alpha parameter, a tuning parameter for controlling the number of decision trees in RF models, on prediction accuracy. We collected the stock price data from the Yahoo Finance website and further preprocessed the data using feature engineering and time series smoothing methods to better fit a model. Although RF is proved to perform well in basic cases, it suffers from a temporal dependency problem considering the time-series data where the relationship between the features is dependent on time. To overcome these disadvantages, RF is proposed to be combined with time series analyses (for instance, ARIMA, GARCH) to handle temporal components within financial data for a specific time period. We evaluate the performance of RF and show that it can handle and generalize the complex relationships between the high-dimensional data. Our experimental analysis underscores the importance of hyperparameter tuning and data preprocessing in enhancing the performance of the RF model and points to imperative areas for integrating the RF with other machine learning algorithms in enhancing financial forecasts.

CCS Concepts

• **Mathematics of computing** → Mathematics analysis; Numerical analysis.

Keywords

Random Forest Model, Alpha Parameter, Model Tuning

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1 Introduction

In the last few decades, data mining and predictive analysis have proven to be essential approaches for many fields, especially for the financial world, where market and security price prediction is important for investment, portfolios, and risks. Historical pattern analysis involves projecting past data in the market with the goal of giving the stakeholders an expectation of what is likely to happen in the market. However, the fluctuating nature of the global financial markets, including their unpredictability and susceptibility to changes, makes the task of accurate prediction quite challenging forresearchers [1]. While baseline statistical models effectively perform with less complex data structures, they are unable to understand the complex curvilinear relationships along with complicated dependency and the high dimensionality that is evident in financial data. These limitations require the use of sophisticated new ML approaches that are well-equipped to identify multi-variate patterns and provide better predictions.

Out of these ML techniques, Random Forest (RF) models are recognized as one of the most promising and applied approaches for prognostic analysis[2]. RF works on an amalgam of decision trees and makes predictions based on the entire set of trees, thus providing higher accuracy and avoiding overfitting. It is very suitable for regression and classification problems, with measures such as the Gini Importance and the Mean Decrease Accuracy (MDA) that give an idea of the most crucial features for the predictive analysis. Fernández-Delgado et al[3]. proved that the RF algorithm is ideal for big data sets and established its irreplaceability in classification tasks.

RF model has some drawbacks, such as they do not model temporal dependencies well, which is the characteristics of any time-series data used in financial forecasts[4]. To overcome these limitations, researchers have suggested blended forecasting using RF in combination with time series methodologies such as Autonomous Recurrent Integrated Moving Average (ARIMA), Generalized Autoregressive Conditional Heteroskedasticity (GARCH), or modern heuristic-based train of thought including support vector machines (SVM) and neural networks.

Other changes made in random forests are guided regularized random forests (GRRF), which was examined by Izquierdo-Verdiguier & Zurita-Milla[5], and considered as the aspect of feature selection in remote sensing. In deriving these, they noted that GRRF uses regularization, which gains 6% classification accuracy and a near 2% regression error reduction, besides eliminating duplicate and irrelevant feature datasets. As Daviran et al.[6] have noted in their work, RF is highly sensitive to hyperparameters like the maximum tree depth as well as the minimum sample splits, so it

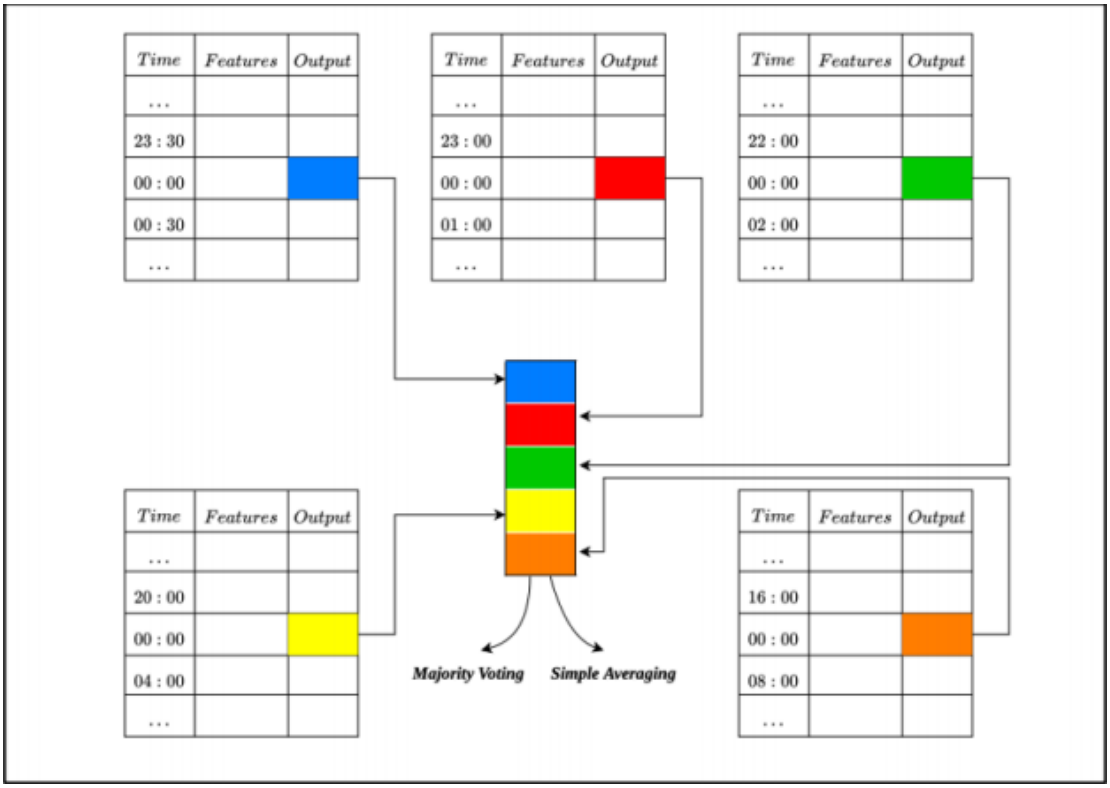


Figure 1: Voting scheme

appears that tuning of hyperparameters may be a critical element in improving the model's robustness and mitigating the case of over-fitting. In this paper, we focus more on the alpha parameter, which determines the number of decision trees in an RF model in the context of financial time-series forecasting. The implications are expected to contribute to the expansion of applying RF to the achievement of financial forecasting by improving it through the integration of such ensemble modeling and hyperparameter tuning.

2 Data

2.1 Data Collection

The analysis used historical financial data taken from Yahoo Finance E-Mini S&P 500 index, in 80 monthly and daily return data, selected from leading firms within the period August 30, 2023 – September 30, 2023. Therefore, to refine the data, the application of the Relative Strength Index (RSI), which is a technical indicator of momentum was adopted to filter overbought/oversold signal data. The momentum indicator applied was the Relative Strength Index (RSI), which helped filter out all the stocks that fell into the overbought or oversold zone. The formula for calculating RSI is as follows –

RSI = Average Gain Over past 80 days/Average Loss Over past 80 days

The average gain was then calculated as was the average loss over an 80-day period and the RSI was obtained using the two averages. The stocks with an RSI value above 70 and below 30 were discarded for the analysis, as the stocks with an RSI>70 represent the overbought situation in the market, and stocks with an RSI<30 determine the oversold situation. The mentioned filtering helped stable market stock movement which positively impacted the model.

In order to observe the daily volatility, the data was arranged on five various timespans including 30 mins, 1 hr., 2 hrs., 4 hrs., and 8 hrs. These intervals corresponded to the closing price at the end of the day at midnight to give a proper framework for studying price changes for several periods at a time.

2.2 Data Pre-processing

The first stage of data preprocessing focuses on enhancing the quality of the data in order to minimize the noise and improve suitability for financial prediction models. These intervals regarding closing price were well-aligned with the daily stock closing price that was obtained from price-to-price extractions at midnight, as shown in Figure 1.

Both intervals took their temporal characteristics into the dataset so that each one of them provided information on the minute changes of stock prices. For simplicity in regression tasks, the last output came out assuming averages on all the intervals in question, while for classification tasks, majority rule voting in boxes was adopted. These methodologies made the robustness of the predictions possible across these different time horizons.

To prepare the bought-in stock data, the exponential smoothing method was used to assign more importance to more recent data compared to older data[7]. This iterative approach can be represented by the following equation –

$$TP_i = Sign\,(CP_{i+d} - CP_i) \tag{1}$$

While the [TP]_i reveals the direction of price changes, the CP renders the price at a particular fixed time period. The [TP]_i values can be either positive or negative, where positive values indicate an increase in the trend and negative values point to a decrease in the trend. This form of preprocessing was especially helpful when it was used to identify short-run market fluctuations.

After data preparation, a daily covariance matrix was computed and applied to analyze the daily and monthly return rates on stocks. The time square root rule was employed in calculating differences between the covariance matrices estimated and the forecasted covariance matrices across the training set with the number of trading days as the time scale. These were carried out to fine-tune the parameters of the model and make the prediction much more accurate. It has been found by Nakach et al.[8] that such a voting scheme provides a rough estimation regarding the intervals in the data and their majority voting and averaging aspects can facilitate the systematic identification in a given case. These preprocessing steps alongside the filtered dataset gave a solid foundation in assessing the Random Forest model in the financial applications.

2.3 Dataset Ensembling

Day volatility is one of the largest deviations from financial market data observed in the market. To effectively reflect these dynamics, the dataset was split into five equal time frames, ranging from 30 minutes to 8 hours. All of the timeframes correspond to the time point which refers to the closing price of the day observed at midnight for conformity between intervals. For instance, an 8-hour digital timeframe often has 3 overlapping Candles aligned to the midnight tick, thus giving an adequate temporal backstop.

The information from these different time frames was incorporated into the Dataset ensembling to increase the reliability of the outputs. Considering classification modeling, majority voting was used where the class that recurred most frequently was taken to be the final prediction. For regression modeling, a straightforward averaging method was usedto average the outputs of all the intervals to give a single comprehensive score. By applying these methods of ensembles, it became possible to avoid the problem of overfitting or bias which lies in the application of a specific interval.

2.4 Data Characteristics

This dataset contains daily and monthly earning statistics from 80 companies. The changes in the price are evaluated by using five intervals of time, which are 30 minutes, 1 hour, 2 hours, 4 hours, and 8 hours. Regarding data preprocessing, filtering was performed by Relative Strength Index (RSI), leaving all data points with RSI>70 and RSI<30 discarded, thus providing more stable input data. Additional aspects of the dataset include time variants and close prices appended with calculated momenta, making it well-structured for classification and regression problems in the prediction of financial rates.

3 Method: Random Forests

The Random Forest (RF) is an overall stable ensemble learning algorithm containing a number of decision trees. Using bagging and feature randomness, RF constructs a set of uncorrelated decision trees, and their collective forecast outperforms the single tree output[9]. The decision tree model is built by partitioning feature spaces into areas where a class node or a leaf node can be used. Then for each tree, a random sample of the features will be used from the dataset to minimize over-fitting giving a higher variance which can be randomly selected for the formation of individual trees considering each split[10]. In order to enhance the performance of the RF classifier, these random variations along with voting-based outcomes will help during the predictive analysis.

In order to create an RF model, there is a standard procedure that is followed. Firstly, there is what is called bootstrap sampling in which several different random samples are made from the original data with replacement. Each subset is then used to train an independent decision tree, in which the decision on splitting the nodes into two groups is carried out based on an optimal feature splitting measure such as the Gini impurity or the mean squared errors. Once all the trees are built, their decisions for a particular record are combined through majority voting for classification problems and averaging for regression problems which makes the model more accurate and reduces variance. It uses an ensemble system of decision trees, which gives it the flexibility to accommodate complex interactions between the predictors, thereby making it suitable for the prediction of financial demands and risks.

4 Experimentation

In order to validate the classifier, the test set data was used with a window size of 80. Measures of cross-validation were used to avert overfitting of the model and to bring the framework into training and test data sets for estimating the consistency of the model. Grid search was carried out to tune hyperparameters such as the alpha denoting the number of decision trees, the maximum depth of decision trees, and the minimum number of samples required to split a branch[11]. These optimizations were intended to increase the generalizability of models with respect to overfitting and underfitting issues.

RF model was compared with conventional methods such as Ridge Regression and Decision Trees alongside with other sophisticated approaches including ARIMA, GARCH, and neural networks. These comparisons gave some indications about how the model was able to generalize and where it failed in terms of temporal dependencies modeling. In the development of the methodology, short pipelines were implemented for experimentation purposes as shown in Figure 2.

The evaluation process requires designing an efficient portfolio using the predicted return rates and the covariance matrices. The benchmark portfolio was obtained by the accumulation of S&P 500 proportions, and the cumulative yield was then recalculated for validation. The Sharpe ratio was employed to analyze performance whereby both means and standard deviations were computed in establishing prediction precision and risk. Further, for scaling over time, the time square root rule was also used to compare the differences in covariances between the predicted and actual one.

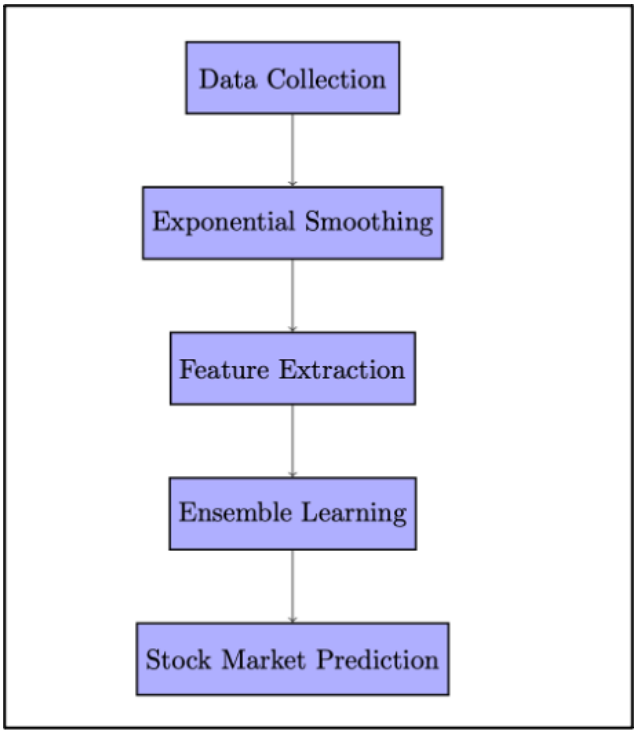


Figure 2: System Architecture of ML-based Pipeline

Other useful factors include technical indicators, such as normalized values of time series training and assessment. The result of the cross-validation will help to showcase how hyperparameters influenced the performance of the model for financial prediction.

5 Result

The random model relies on the features to capture time-based trends in the data, as it cannot see lagging variables. To crack Random Forest processing time series data, an ML-based pipeline was created, which includes expected returns, expected draw returns, actual returns, and their variances. The performance statistics of the classifier are presented in Table 1, which shows the Sharpe ratio variations of parameters.

In the context of validating the preprocessing methods, the identified parameters of returns, including the Yearly Rate Difference (YRD) and Sharpe ratios, were compared to real data. Interestingly, the Sharpe differences ranged between -2.3390 and -3.6252, with the Standard Deviations ranging from 5.7452 to 6.8091. The outcome of the analysis indicates areas for improvement and opportunities for a combination of techniques to increase the efficiency and reliability of the models in financial forecasting.

The advantage of a Random Forest is its robustness to a large number of variables. Therefore, collinearity is the only major issue when selecting which technical indicators to include in the model. We trained several Random Forests and selected variables based on the combination that provided the highest accuracy on the test set. Figure 3 represents the performance of the Random Forest model.

Table 1 determines that results with the highest number of parameters show the metrics for the Random Forest model in predicting cumulative returns and Sharpe ratios. The left subplot from Figure 3 further shows how the model was able to identify better trends in the predicted portfolio returns that were above the benchmark index. However, the differences indicate that prediction disparities are more apparent during periods of volatility, following the increasing trends in the Sharpe ratios as observed in the result table[12]. The middle subplot from Figure 3 shows the relative measures of cumulative returns estimated by the model and the cumulative actual returns on a sample index, which are consistent considering the difference as 0, implying the model can effectively capture temporal structure and evolving characteristics of markets.

The right plot graph of Sharpe ratio comparisons exhibits significant fluctuations in the test Sharpe ratio commensurate with the standard deviation range (6.1632 to 6.8091) presented in Table 1. The negative mean Sharpe differences varying from -2.339 to -3.5973show the realization of persistent prediction errors under any conditions. These results imply further optimization and integration to improve the reliability and robustness of financial forecasting.

6 Conclusion

The financial market is rather noisy and constantly evolving, making it challenging to develop models to predict future outcomes, especially in terms of time dependencies. Although the RF model presented good generality and efficiency in handling high-dimensional data, its application was not efficient in dealing with temporal dependencies of Forecasting Financial Series. Considering the values of the “Mean Sharpe Difference” and “SD of Sharpe Difference” given by the model presented in Table 1, it became clear that the alpha parameter is sensitive and can be adjusted for efficiency, but it was unable to deliver enough accuracy.

These limitations can be addressed, for which it is recommended that RF should be integrated with time series models like ARIMA or GARCH during the analysis of sequential data. However, combining RF with other complicated and higher-level algorithms like support vector machines or neural networks may further improve its prediction capability and generality.

The experimental analysis showed that increasing the number of trees and optimizing the Random Forest model indeed improved the performance in the chosen criteria. The performance index was greater than the benchmark index, and the Sharpe ratio changes were also indicative of improved risk-adjusted return rates. Furthermore, the optimized model will capture more complex relationships to reduce the bias so that the financial forecasting performance and veracity in unpredictable markets can be enhanced.

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Table 1: Random Forest – Performance statistics

Parameter	YRD with Baseline	YRD with Real Data	Mean Sharpe Difference	SD of Sharpe Difference
1	16.19378	0	-2.339042	6.8091
2	23.823003	0	-3.170278	6.634428
3	18.342427	0	-3.046388	6.494653
4	20.795078	0	-3.151348	6.606066
5	25.797174	0	-3.416073	6.664849
6	27.669191	0	-3.514456	6.663295
7	28.751189	0	-3.626199	6.632173
8	26.581663	0	-3.420155	6.390342
9	26.118494	0	-3.391694	5.928663
10	26.440715	0	-3.402831	5.745212
11	25.833955	0	-3.430228	5.807431
12	25.419785	0	-3.437818	5.919895
13	26.472263	0	-3.53157	5.870447
14	26.674608	0	-3.553577	5.893595
15	25.79524	0	-3.475788	5.930334
16	27.177453	0	-3.56402	5.979138
17	28.678946	0	-3.625197	5.934996
18	28.619611	0	-3.61556	6.021973
19	27.344488	0	-3.597345	6.163231

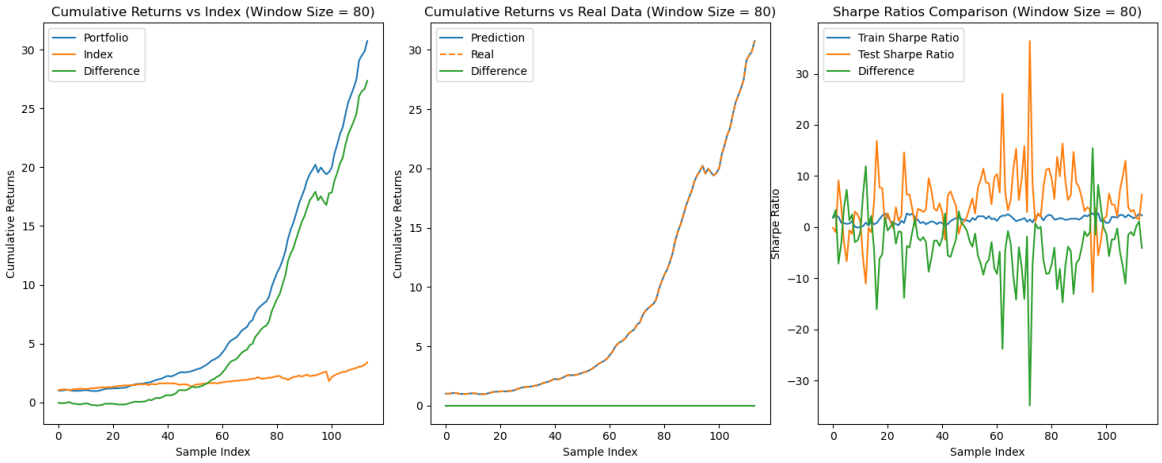


Figure 3: Random Forest Performance Analysis.

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